

CLI BANK PROJECT



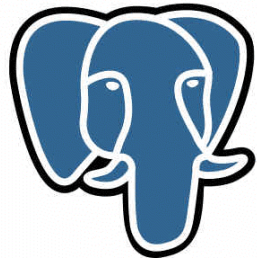
SYLVIE SPLAVER



Tech Stack

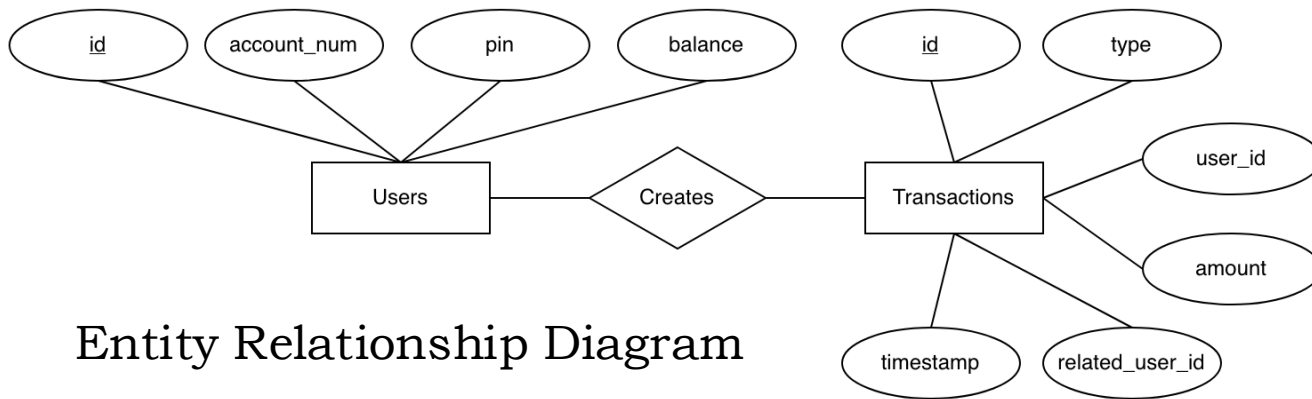


PostgreSQL



- Primary Language: **Java**
- Version Control: **Git + Github**
- Database: **PostgreSQL**
- Testing Framework: **JUnit 6**
- Build Tool & Dependency Management: **Maven**

Design



Entity Relationship Diagram

- Multi-layered Architecture
 - ❖ API
 - ❖ Service/Business
 - ❖ Repository
- OOP Principals
 - ❖ APIE
- Heavy user input validation
 - ❖ Cover the majority edge cases

Visual Design

```
[Account]
Balance: $117,056.00 | Account #: 100009

Logged into your Bank of CLI account!
```

```
[Welcome]
Please login to see account information.

Welcome to the Bank of CLI!

Use the "help" command to see available commands
```

Dedicated **CLIFormatter** Object

- Dynamically reads data from .json file
- Generates a UI box to contain the text
- Supports Text Justification: Left, Right, & Center
- Supports Title and Page number generation
- Supports custom keywords to make repetitive typing easier

Unit Testing

- Focused on Positive and Negative Tests in the Business/Service layer
- Using Mockito to simulate account retrieval



✓ AuthenticationServiceTest (com.sylvie.clibank.business)	1 sec 194 ms
✓ signInPositive()	424 ms
✓ signInNegative_WrongPin()	2 ms
✓ signInNegative_InvalidAccountNum()	2 ms

Example: Positive and Negative tests related to the signIn() method

Logging

- Embedded across the Business/Service layer
- Logs are written to files in the logs folder
 - ❖ INFO for key events for the bank
 - ❖ ERROR for invalid user inputs or runtime issues

```
2026-09-23 14:32:34 [main] INFO c.s.clibank.business.UserService - Account # 100009 Deposited: 10000.0
2026-09-23 15:26:48 [main] ERROR c.s.clibank.business.UserService - Account # 100009 tried to deposit a negative amount!
2026-09-23 15:26:53 [main] INFO c.s.clibank.business.UserService - Account # 100009 Deposited: 2000.0
2026-09-23 15:27:42 [main] INFO c.s.clibank.business.UserService - Account # 100009 Withdrew: 99.0
2026-09-23 15:27:48 [main] ERROR c.s.clibank.business.UserService - Account # 100009 tried to withdraw a negative amount!
2026-09-23 15:27:54 [main] INFO c.s.clibank.business.UserService - Account # 100009 Withdrew: 0.0
```

DEMO

